

Fast Parallel Image Rotation Algorithm

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Abstract—This paper presents a novel fast image rotation algorithm that leverages CPU parallelization while maintaining high image quality. The proposed method is a single-pass algorithm, derived from the existing Double-Line Rotation (DLR) method, which reduces computational complexity and generalizes the algorithm. Initially, a baseline for the given image is calculated to determine the starting line, which defines the initial point for each vertical or horizontal line in the image to be rotated. The corresponding pixels to be mapped are identified using floating point arithmetic, and trigonometric calculations are performed only once per line. This approach ensures precise image transformation with minimal computational overhead.
Index terms: Image rotation, line rotation image transform, double-line rotation (DLR), parallel image rotation.

I. INTRODUCTION

2-D Image Rotation has a number of applications in digital photography and image processing like image matching, alignment, [1] etc. Highly accurate image rotation is essential for certain image processing tasks such as feature extraction and matching. Whilst there are many state-of-the-art image rotation algorithms, they have limitations in terms of image quality and speed [2] [3]. Parallel algorithms for image processing aim to execute faster than sequential algorithms, however designing and scheduling a fast parallel image rotation is a challenge [4]. The most straightforward, and perhaps the most well known approach to implement image rotation is by using the rotation matrix [5], as described here

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x - cx \\ y - cy \end{bmatrix} * \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

where, θ is the angle of rotation, x' and y' are the new

coordinates of the rotated image, and (cx, cy) is the center of the original image, which is treated as the axis of rotation. Assuming the centre of rotation of the new image is (cx', cy') , the final new coordinates are:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x' + cx' \\ y' + cy' \end{bmatrix}$$

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II. LITERATURE SURVEY

A. Affine Transformation

A basic idea in computer graphics and geometry, affine transformation is a linear mapping method that maintains straight lines, planes, and points. The capacity to carry out a variety of geometric operations, including translation, scaling, rotation, shearing, and reflection, is what defines an Affine transformation. These transformations are powerful and adaptable tools with a wide range of applications because they are defined mathematically by combining translations and linear transformations.

1) Properties:

- **Linearity and Translation:** The transformation is composed of a linear transformation Ax and a translation β . This combination allows for a wide range of geometric transformations while preserving the structure of the space.

- Preservation of Collinearity and Ratios: Affine transformations preserve the collinearity of points and the ratios of distances along parallel lines.
- Combining Transformations: Multiple affine transformations can be combined into a single affine transformation by matrix multiplication and vector addition.

MATHEMATICAL FORMULATION

Affine transformations can be described by a combination of linear transformations and translations. The general form of an affine transformation in two dimensions is:

$$y = Ax + b$$

where

- $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ is the input coordinate vector.
- $\mathbf{A}$ is a 2×2 matrix that represents the linear part of the transformation.
- $\mathbf{b} = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$ is the translation vector.

For a rotation by an angle θ around the origin, the matrix $\mathbf{A}$ is:

$$\mathbf{A} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

To center the rotated image within its original dimensions, the translation values t_x and t_y are calculated as follows:

CALCULATIONS

New Dimensions

The new width and height of the rotated image are determined using trigonometric functions based on the rotation angle:

$$\text{newWidth} = \lceil \text{width} \cdot |\cos(\theta)| + \text{height} \cdot |\sin(\theta)| \rceil$$

$$\text{newHeight} = \lceil \text{width} \cdot |\sin(\theta)| + \text{height} \cdot |\cos(\theta)| \rceil$$

Center Points

fine transformation matrix that includes rotation and translation is:

$$\mathbf{T} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \end{bmatrix}$$

The new coordinates (x', y') after applying the affine transformation are:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = \cos(\theta) \cdot x - \sin(\theta) \cdot y + t_x$$

$$y' = \sin(\theta) \cdot x + \cos(\theta) \cdot y + t_y$$

$$c_x = \frac{\text{width}}{2}, \quad c_y = \frac{\text{height}}{2}$$

$$\text{newCx} = \frac{\text{newWidth}}{2}, \quad \text{newCy} = \frac{\text{newHeight}}{2}$$

Translation Values

The translation values to center the rotated image within its new dimensions are:

$$t_x = c_x - (\text{newCx} \cdot \cos(\theta) - \text{newCy} \cdot \sin(\theta))$$

$$t_y = c_y - (\text{newCx} \cdot \sin(\theta) + \text{newCy} \cdot \cos(\theta))$$

Thus, the affine transformation matrix that includes rotation and translation is:

$$\mathbf{T} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \end{bmatrix}$$

The new coordinates (x', y') after applying the affine transformation are:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = \cos(\theta) \cdot x - \sin(\theta) \cdot y + t_x$$

$$y' = \sin(\theta) \cdot x + \cos(\theta) \cdot y + t_y$$

III. PROPOSED METHOD

The proposed method is a single pass algorithm, which means that it performs the entire image rotation operation at once. The proposed algorithm predominantly affects pixels in the row major (width), or ones in the column major (height), depending on the angle of rotation, α . If $\alpha \in (\frac{\pi}{4}, \frac{3\pi}{4}] \cup (\frac{5\pi}{4}, \frac{7\pi}{4}]$, rotation is performed in the **column major**, and it is performed in the **row major** for the rest of the angles. Accordingly, for rotating the source image, entire row or column vectors of pixels are rotated as a single unit, instead of the more common approach, where each individual pixel is separately mapped. Due to this reason, it is necessary to accurately determine the initial coordinates from which the new row or column vector will begin. The coordinates of the rest of the pixels in the vector can then be determined by simple floating point addition or subtraction, depending on the extent to which each of the said vectors are spread across the coordinate axes.

Apart from this, depending on α , the pixel processing order changes - i.e., whether the last row or column vector of pixels get processed first, or last. Since it has been found out that these values are very sensitive to α , the specific details of which are discussed in the later parts of this paper.

A. Rotation Of An Image In The Column Major

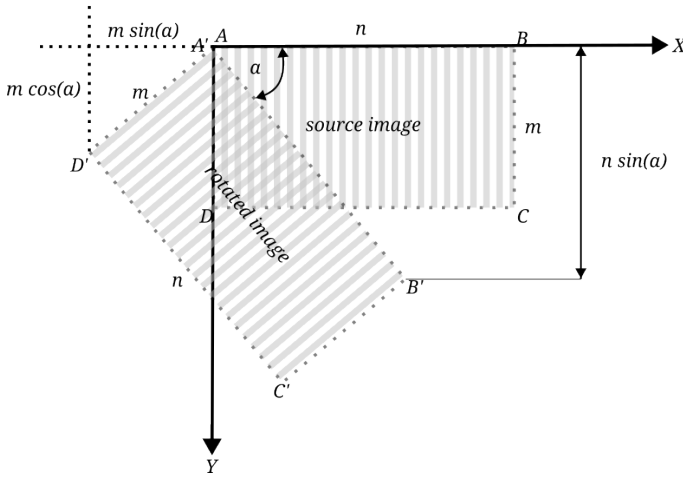


Fig. 1: Example demonstrating rotation of an image in the column major. Every column vector of pixels are rotated at once (Represented by the stripes in the image).

As seen in the example image in figure 1, for the given angle of rotation, column vectors were rotated (represented by the stripes). Each such vector to be rotated begins from the line $\overline{A'B'}$. As such, the initial coordinates of each of these vectors can be determined by the equation of the same line, which is given by:

$$f(i) = \left\lfloor \frac{i}{\tan(\alpha)} \right\rfloor \forall \{i \in \mathbb{Z}, i \in [0, n \cdot \sin(\alpha))\} \quad (1)$$

Thus, the initial coordinates of the column vectors would be of the type $(f(i), i)$. Next, to determine the coordinates of the rest of the pixels in the column vector, as mentioned previously, we perform floating point addition or subtraction, depending on the extent to which each of the said vectors are spread across the coordinate axes.

In the example in figure 1, we can see that each column vector is spread through a length of $m \cdot \sin(\alpha)$ in the x -axis and through a length of $m \cdot \cos(\alpha)$ in the y -axis. It may also be observed that these values are the projections of any line perpendicular to line defined by the function in equation 1 on the coordinate axes.

We now define:

$$\delta x = |\sin(\alpha)|$$

$$\delta y = |\cos(\alpha)|$$

The coordinates of the next pixel in the column vector is then given by:

$$(f(i) + \delta x, i + \delta y)$$

The coordinates of the pixel following that is determined by adding δx and δy to the previous coordinates accordingly, and so on, for each of the m pixels in the given column vector. The end-result is that this operation covers all the $m \cdot \sin(\alpha)$

pixels in the x -axis and all the $m \cdot \cos(\alpha)$ pixels in the y -axis, therefore mapping all of the m pixels in the specific column vector.

This process continues for all of the n column vectors in the image, until the entire image is rotated. In practice, each column vector is actually mapped **twice**. The first mapping happens from the i th column vector in I to the i th column vector in R . The second mapping happens from the i th column vector in I to the $(i+1)$ th column vector in R . Each of these $(i+1)$ th column vectors get overwritten in each iteration with a new column vector in I ; however, this gives a sufficiently accurate issuance of anti-aliasing without losing much quality.

The signs for δx and δy depend on whether, while mapping the column vectors in the rotated image, the values of the x -coordinate y -coordinate are decreasing or increasing. In the given example in figure 1, δx is negative, but δy is positive. There is a third parameter, defined as:

$$\delta i = \frac{1}{\delta x}$$

δi determines the rate at which the column vector is processed from one to the next, and its sign depends on the angle of rotation, α . The calculations for δx , δy , and δi only happen once per column vector in the image, so the entire image transformation is quite fast, overall.

B. Rotation Of An Image In The Row Major

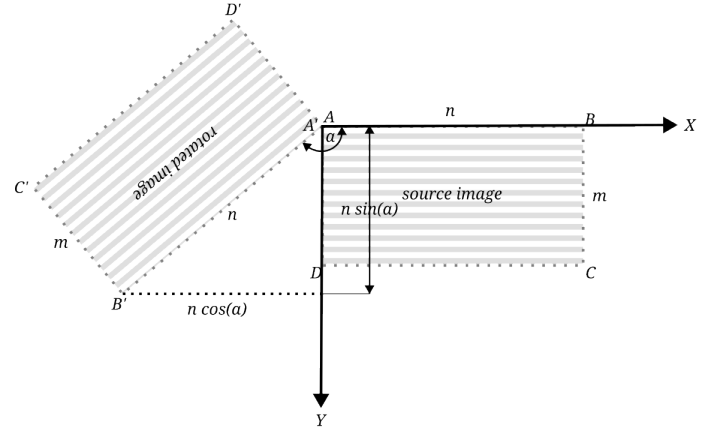


Fig. 2: Example demonstrating rotation of an image in the row major. Every row vector of pixels are rotated at once (Represented by the stripes in the image).

In the example image in figure 2, for the given angle of rotation, row vectors were rotated (represented by the stripes). All the calculations and logic is very similar to the one described for rotation in the column major [III-A]. Here, each vector to be rotated begins from the line $\overline{A'D'}$, whose equation is given by:

$$f(i) = \left\lfloor \frac{i}{\tan(\alpha + \frac{\pi}{2})} \right\rfloor \forall \{i \in \mathbb{Z}, i \in [0, m \cdot \cos(\alpha))\} \quad (2)$$

TABLE I: Notation

Symbol	Description
I	Original image
m, n	Height, width of I
R	Rotated image
m', n'	Height, width of R
α	Angle of rotation in radians
x'	Offset by which the x -coordinates of R need to be shifted
y'	Offset by which the y -coordinates of R need to be shifted
$f(\dots)$	Function defining the equation of the line from where the row or column vector starts [eq. 1] [eq. 2]
r_o	Upper limit of the function f
N	Number of elements (pixels) in the row or column vector f
flip	Boolean value that changes pixel processing order based on α
row_major	Boolean value that determines whether processing is to be done in the row or column major
negate	Boolean value, that if true, negates the indices used for indexing the pixels of the image

Note: The algorithm described in [Algorithm 1] follows *Python* style indexing. For example, if $a = [[1, 2, 3], [5, 6]]$, then $a[0][1]$ represents the element 3.

Everything else stays the same, except the fact that now, each row vector is spread through a length of $n \cdot \cos(\alpha)$ in the x -axis and through a length of $n \cdot \sin(\alpha)$ in the y -axis, and accordingly, we have:

$$\delta x = |\cos(\alpha)|$$

$$\delta y = |\sin(\alpha)|$$

C. Implementation

Algorithm 1 Image Rotation Algorithm

```

1: procedure ROTATE(...)  $\triangleright$  Performs image rotation on  $R$ 
2:    $\text{src\_i} = 0$   $\triangleright$  Initialize internal control variables
3:    $S = 1$ 
4:    $M = []$ 
5:   if negate then
6:      $S = -1$ 
7:   end if
8:   if flip then
9:      $M = [m_i]$  for  $i = 0, 1, \dots, N - 1$ ,  $| m_i = N - i$ 
10:  else
11:     $M = [m_i]$  for  $i = 0, 1, \dots, N - 1$ ,  $| m_i = N$ 
12:  end if
13:  for  $i = 0$  to  $r_o$  do
14:     $\text{temp\_x} = f(i)$ 
15:     $\text{temp\_y} = i$ 
16:    for  $px$  in  $M$  do
17:       $x = \lfloor \text{temp\_x} \rfloor$ 
18:       $y = \lfloor \text{temp\_y} \rfloor$ 
19:       $\text{temp\_i} = \lfloor \text{src\_i} \rfloor$ 
20:      if row_major then
21:         $a = \text{temp\_i}$ 
22:         $b = px$ 
23:      else
```

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24:         $a = px$ 
25:         $b = \text{temp\_i}$ 
26:      end if
27:       $R[\lfloor S * (y + y') \rfloor, \lfloor S * (x + x') \rfloor] = I[a, b]$ 
28:       $R[\lfloor S * (y + y') + 1 \rfloor, \lfloor S * (x + x') \rfloor] = I[a, b]$ 
29:       $\text{temp\_x} += \delta x$ 
30:       $\text{temp\_y} += \delta y$ 
31:    end for
32:     $\text{src\_i} += \delta i$ 
33:  end for
34: end procedure
```

The algorithm described above [Algorithm 1] rotates image I to R .

R is Initialized as a blank image of width n' and height m' to fit R in the canvas, where:

$$m' = \lceil |m \cdot \cos(\alpha) + n \cdot \sin(\alpha)| \rceil$$

$$n' = \lceil |m \cdot \sin(\alpha) + n \cdot \cos(\alpha)| \rceil$$

The rotation function is then invoked, with each of the parameters updating depending on the value of α , as discussed above. The detailed values of each of the parameters are mentioned in [Table II].

Note: It has been observed, that for ranges of α which are vertically opposite each other (For example, angles in the range $(\frac{\pi}{4}, \frac{\pi}{2}]$ are vertically opposite angles in the range $(\frac{5\pi}{4}, \frac{3\pi}{2}]$, and so on), the only parameters of rotation that change are δi , negate, and flip. As such, the calculations have been generalized for such angles and they have been represented in [Table II], with a (') at the end of their names.

Algorithm 2 Driver Function

```

1: procedure MAIN(...)  $\triangleright$  Invokes [Algorithm 1]
2:   Determine the boolean value of row_major
3:   Determine  $f$ , based on the value of row_major
4:   Initialize a blank image  $R$  with dimensions  $(m', n')$ 
5:   if  $0 \leq \alpha \leq \frac{\pi}{4}$  then
6:      $x' = \lceil |m \cdot \sin(\alpha)| \rceil$ 
7:      $y' = 0$ 
8:      $\delta x = |\cos(\alpha)|$ 
9:      $\delta y = |\sin(\alpha)|$ 
10:     $\delta i = \frac{1}{\delta x}$ 
11:     $r_o = \lceil |m \cdot \cos(\alpha)| \rceil$ 
12:     $N = n$ 
13:    negate = False
14:    flip = False
15:  else if  $\frac{\pi}{4} < \alpha \leq \frac{\pi}{2}$  then
16:     $x' = \lceil |m \cdot \sin(\alpha)| \rceil$ 
17:     $y' = 0$ 
18:     $\delta x = -|\sin(\alpha)|$ 
19:     $\delta y = |\cos(\alpha)|$ 
20:     $\delta i = \frac{1}{\delta x}$ 
21:     $r_o = \lceil |n \cdot \sin(\alpha)| \rceil$ 
```

TABLE II: Rotation Parameters, based on α

$\frac{3\pi}{4}, \pi$	$\left(\frac{3\pi}{2}, \frac{3\pi}{4}\right)$	$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$	$[0, \frac{\pi}{4}]$	Range of α	Rotation parameters									
					m'	$[[m \cdot \sin(\alpha)]]$	$[[m \cdot \sin(\alpha)]]$	x'	y'	δx	δy	δi	r_o	$\mathbf{Z}$
$[[n \cdot \cos(\alpha)]]$	0	$[[m \cdot \cos(\alpha)]]$	0		$[[m \cdot \cos(\alpha)]]$	0	$[[m \cdot \sin(\alpha)]]$	$[[m \cdot \sin(\alpha)]]$	0	$ \cos(\alpha) $	$ \sin(\alpha) $	$1/\delta x$	$[[m \cdot \cos(\alpha)]]$	n
$- \cos(\alpha) $	$- \sin(\alpha) $	$- \sin(\alpha) $	$- \sin(\alpha) $		$- \sin(\alpha) $	$- \sin(\alpha) $	$ \cos(\alpha) $	$ \cos(\alpha) $						false
$ \sin(\alpha) $	$- \cos(\alpha) $	$- \cos(\alpha) $	$ \cos(\alpha) $		$- \cos(\alpha) $	$ \cos(\alpha) $	$ \sin(\alpha) $	$ \sin(\alpha) $						false
$1/\delta x$	$-1/\delta x$	$-1/\delta x$	$1/\delta x$		$-1/\delta x$	$1/\delta x$	$1/\delta x$	$1/\delta x$						flip
$[[m \cdot \cos(\alpha)]]$	$[[n \cdot \sin(\alpha)]]$	$[[n \cdot \sin(\alpha)]]$	$[[n \cdot \sin(\alpha)]]$		$[[n \cdot \sin(\alpha)]]$	$[[n \cdot \sin(\alpha)]]$	$[[m \cdot \cos(\alpha)]]$	r_o						negate
n	m	m	n		m	m	n	$\mathbf{Z}$						flip
false	false	false	false		false	false	false	negate						
false	false	false	false		false	false	false	flip						
$1/\delta x$	$1/\delta x$	$1/\delta x$	$1/\delta x$		$1/\delta x$	$1/\delta x$	$1/\delta x$	$\delta i'$						
true	false	false	true		false	false	true	negate'						
false	true	true	false		true	true	false	flip'						

```

22:      $N = m$ 
23:     negate = False
24:     flip = False
25: else
26:      $a = px$ 
27:      $b = \text{temp\_i}$ 
28: end if
29:     Generate  $R$  using [Algorithm 1]
30: end procedure

```

IV. CONCLUSION

The conclusion goes here.

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